Volatility clustering, fat tails and Value-at-Risk for the Nikkei 225

Gaussian, Merton jump-diffusion and GARCH one-day VaR, trained on 2000–2011 and backtested on 2012–2026

1 Question and data

Question. Can a one-day Value-at-Risk (VaR) model estimated on 2000–2011 Nikkei 225 data pass standard coverage and independence backtests on 2012–2026? And which feature of the returns matters more for that: fat tails (Merton jump-diffusion) or volatility clustering (GARCH)?

Data. Daily closing levels of the Nikkei 225 (Yahoo Finance ticker `^N225`, auto-adjusted), downloaded with `yfinance` on 30 October 2026 and stored as `data/n225.csv`. Prices run from 4 January 2000 to 30 October 2026, giving 6,569 daily log returns in percent,

$$r_t = 100 \ln(P_t/P_{t-1}) .$$

The training set is every return before 1 January 2012 (2,943 days, 5 Jan 2000 – 30 Dec 2011). The test set is every return from then on (3,626 days, 4 Jan 2012 – 30 Oct 2026). Returns are fat-tailed (kurtosis 9.8, skewness -0.38). On the training data the Ljung–Box test finds little autocorrelation in the returns themselves ($p = 0.15$ at 5 lags, 0.11 at 10). It finds very strong autocorrelation in squared returns ($Q = 1,484$ at 5 lags, $p < 0.001$ at every lag tested), so volatility clusters.

2 Methods

Mean forecasts (baseline). OLS autoregressions with 1–5 lags, fitted with scikit-learn's `LinearRegression`. They are refitted every test day on all earlier days (expanding walk-forward, so the forecast for day t uses data up to $t-1$ only). They are compared with a zero forecast and with the expanding historical mean, using out-of-sample R^2 and a simple t-statistic on the daily squared-error difference, without a correction for autocorrelation.

Gaussian. Returns are i.i.d. $\mathcal{N}(\mu, \sigma^2)$, with maximum-likelihood estimates from the training data.

Merton jump-diffusion. Returns are i.i.d. with $r = \mu + \sigma Z + \sum_{j=1}^{N} J_j$, where $N \sim \text{Poisson}(\lambda)$ and $J_j \sim \mathcal{N}(m, \delta^2)$. This gives a Poisson-weighted mixture of normals, truncated at $K = 10$ jumps:

$$f(r) = \sum_{k=0}^{K} \frac{e^{-\lambda}\lambda^k}{k!}\, \phi\big(r;\ \mu + km,\ \sigma^2 + k\delta^2\big) .$$

I wrote the likelihood by hand and maximised it with scipy's Nelder–Mead from three starting points; all three reach the same log-likelihood. The two densities are compared out of sample by the daily log-score difference

$$d_t = \log f_{\text{Merton}}(r_t) - \log f_{\text{Gauss}}(r_t),$$

decomposed by $|r_t|$ bucket. A moving-block bootstrap gives a confidence interval for its mean (5,000 resamples, blocks of 10, 20 and 60 days, seed 42).

GARCH. With $\varepsilon_t = r_t - \mu$ (μ is the training mean), the conditional variance follows

$$\sigma_t^2 = \omega + \big(\alpha + \gamma\, \mathbf{1}[\varepsilon_{t-1} < 0]\big)\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2, \qquad \varepsilon_t = \sigma_t z_t.$$

- **Own MLE.** The Gaussian GARCH(1,1) ($\gamma = 0$) is fitted by my own maximum-likelihood code, using scipy SLSQP with $\alpha + \beta < 1$, on the log-likelihood $-\frac{1}{2}\sum_t \big[\ln 2\pi + \ln \sigma_t^2 + \varepsilon_t^2/\sigma_t^2\big]$.
- **Check against `arch`.** With the same start-up variance the two fits agree to six decimal places ($\omega = 0.0438$, $\alpha = 0.1092$, $\beta = 0.8760$, log-likelihood -5172.71).
- **Scaling.** Returns are scaled by 100 because the optimiser stops early on decimal returns.
- **Model choice.** Student-t innovations and the GJR leverage term γ were then added one at a time, using `arch`. The backtest model was chosen on training BIC before any backtest was run. The winner was GJR-t: $\omega = 0.047$, $\alpha = 0.025$, $\gamma = 0.123$, $\beta = 0.894$, $\nu = 14.3$. Its persistence $\alpha + \gamma/2 + \beta = 0.980$ gives a half-life of $\ln 0.5/\ln(0.980) \approx 34$ trading days.

VaR and backtests. One-day VaR at tail probability $a \in \{5\%, 1\%, 0.5\%\}$ (95%, 99% and 99.5% VaR), reported as a positive loss in percent:

$$\text{VaR}^{\text{Gauss}} = -(\mu + \sigma\, \Phi^{-1}(a)), \qquad F_{\text{Merton}}\Big(-\text{VaR}^{\text{Merton}}\Big) = a, \qquad \text{VaR}_t^{\text{GJR-}t} = -\Big(\mu + \sigma_t\, t_\nu^{-1}(a)\sqrt{\tfrac{\nu-2}{\nu}}\Big).$$

The Merton quantile is found numerically with Brent's method; the square root rescales the t distribution to unit variance.

- **Fixed parameters.** The same scheme is used for every model: parameters are estimated once on 2000–2011 and never refitted. Gaussian and Merton VaR are therefore constant.
- **Daily GARCH update.** GJR-t VaR still moves because σ_t^2 is updated each day by the recursion, using returns up to $t-1$ only. This is checked in code: changing the last return leaves every forecast unchanged.
- **Tests.** A breach is a day with $r_t < -\mathrm{VaR}_t$. With x breaches in N days, $\hat{\pi} = x/N$, and n_{ij} the number of transitions from state i to state j:

$$LR_{uc} = -2\ln\frac{(1-a)^{N-x}a^x}{(1-\hat{\pi})^{N-x}\hat{\pi}^x}, \qquad LR_{ind} = -2\ln\frac{(1-\tilde{\pi})^{n_{00}+n_{10}}\tilde{\pi}^{\,n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}}\hat{\pi}_{01}^{n_{01}}(1-\hat{\pi}_{11})^{n_{10}}\hat{\pi}_{11}^{n_{11}}}.$$

Here $\hat{\pi}_{01} = n_{01}/(n_{00}+n_{01})$ and $\hat{\pi}_{11} = n_{11}/(n_{10}+n_{11})$ are the breach probabilities after a quiet day and after a breach, and $\tilde{\pi} = (n_{01}+n_{11})/(N-1)$ is the overall breach rate over the transitions. These are Kupiec's unconditional coverage test ($LR_{uc} \sim \chi^2_1$), Christoffersen's first-order Markov independence test ($LR_{ind} \sim \chi^2_1$) and their sum $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2_2$.

3 Results

Mean forecasts. No lag order beats the naive forecasts in a meaningful way. Out-of-sample R^2 is at most 0.0012 against zero and 0.0014 against the historical mean, and every $|t|$ is below 0.75. Daily returns are close to unpredictable in the mean, so VaR is driven by the shape and scale of the distribution.

Merton vs Gaussian density. Merton fits better both in sample and out of sample.

- **In sample:** the training likelihood-ratio statistic is 415 for three extra parameters. The fitted values are $\sigma = 1.26\%$, $\lambda = 0.096$ jumps per day, mean jump $m = -0.48\%$ and jump sd $\delta = 3.13\%$.
- **Out of sample:** the mean log-score gain is $\bar{d} = 0.077$ per day (block-bootstrap 95% CI 0.056 to 0.104 across block lengths).
- **Where the gain comes from:**
  - Ordinary days with $|r| < 1\%$ (2,333 days) contribute 128% of the total gain, because Merton's diffusion part is narrower and puts more density near zero.
  - Days with $|r|$ between 2% and 3% *reduce* the gain by 40%.
  - The 128 days with $|r| \geq 3\%$ add 24%.

The better density is therefore mainly a better centre, not better tails.

Table 1: Out-of-sample VaR backtests, 3,626 test days. p-values in bold reject the null at 5%.

Model	VaR level	Breaches (expected)	Kupiec p	Christoffersen p	LR_{cc} p
Gaussian	95%	110 (181)	**<0.001**	**<0.001**	**<0.001**
	99%	40 (36)	0.539	**0.010**	**0.029**
	99.5%	25 (18)	0.126	**0.011**	**0.012**
Merton	95%	133 (181)	**<0.001**	**<0.001**	**<0.001**
	99%	17 (36)	**<0.001**	0.068	**<0.001**
	99.5%	6 (18)	**<0.001**	0.888	**0.004**
GJR-t	95%	176 (181)	0.685	0.610	0.808
	99%	53 (36)	**0.009**	0.237	**0.016**
	99.5%	30 (18)	**0.011**	0.250	**0.020**

VaR backtests (Table 1).

- **Gaussian:** about the right number of 99% breaches (40 against 36 expected), but they cluster: Christoffersen $p = 0.0095$.
- **Merton:** its heavier left tail makes its VaR too conservative: 17 breaches at 99% and 6 at 99.5%, rejected by Kupiec at both. It passes the independence test at 99.5% only because it has so few breaches.
- **GJR-t, clustering:** fixed. Its standardised residuals show no remaining autocorrelation in z_t^2 (Ljung–Box $p \geq 0.31$ at 5, 10, 20 and 40 lags, against $p < 0.001$ for raw squared returns). The Christoffersen test passes at every level ($p = 0.24$ at 99%), and at 95% the model passes all three tests.

- **GJR-t, tail coverage:** at 99% and 99.5% it breaches too often (53 and 30, against 36 and 18 expected), so Kupiec and LR_{cc} reject. Residual kurtosis falls from 9.5 for raw training returns to 3.9. That is still above the $3(\nu-2)/(\nu-4) = 3.6$ implied by t_{14}, so the far tail is still too thin.

The result is mixed. GARCH removes the clustering that sank the Gaussian model, but at the 99% level it trades the Gaussian's correct breach count for too many breaches.

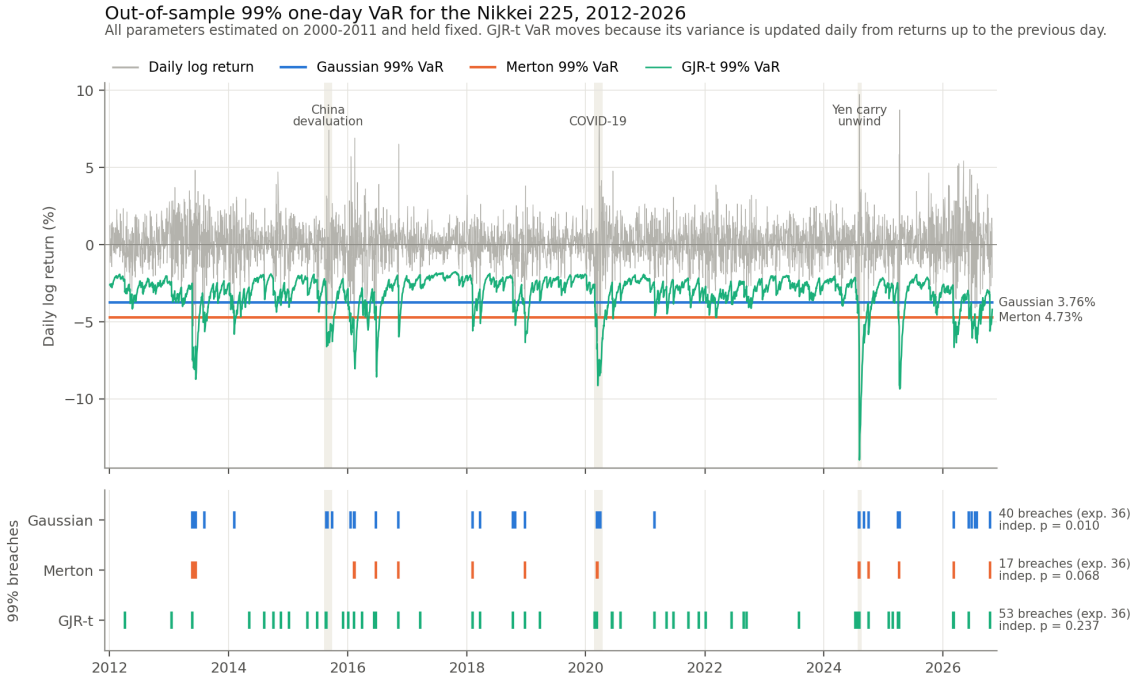


Figure 1: Test-period returns with each model's 99% VaR (top) and the dates of 99% breaches (bottom). Shaded: August–September 2015, February–April 2020 and late July–August 2024. Under GJR-t, breaches no longer follow one another day after day, but crisis years still have more than their share: 6, 5 and 5 breaches in 2015, 2020 and 2024, against about 2.4 expected per year. This is because the variance only rises after the first large fall.

4 Limitations

All parameters come from a single 2000–2011 estimation and are never refitted. The test period is calmer than training and has a higher mean (daily sd 1.37% vs 1.60%, mean +0.056% vs −0.028%). Some of GJR-t's excess 99% breaches may reflect stale parameters rather than the model form; an expanding-window refit was not tested. There is one index, one split and one data source (Yahoo Finance, not cross-checked against an official series). Six VaR models at three levels are tested, so a few rejections or passes are expected by chance, and the tests are not adjusted for this. The model was chosen on training BIC. GARCH-t, which was not chosen, did better ex post at 99% (48 breaches, LR_{cc} $p = 0.066$), but switching to it now would be selecting on the test set. Christoffersen's test only detects day-to-day dependence, so it can pass while breaches still bunch within weeks, as in Figure 1. Ljung–Box p-values on GARCH residuals use h rather than $h-2$ degrees of freedom, the OLS t-statistics ignore autocorrelation and heteroskedasticity, and the Merton density is truncated at 10 jumps.

5 Conclusion

Daily Nikkei returns are almost unpredictable in the mean, so VaR quality depends on the distribution.

- **Fat tails alone (Merton)** improve the density fit, mainly near the centre, but give a VaR that is too conservative and still constant.
- **Volatility clustering** is the feature that fixes the Gaussian model's failure. GJR-t raises the 99% Christoffersen p from 0.0095 to 0.24 and passes every test at 95%.
- **The extreme tail is not fixed.** With parameters fixed at 2000–2011 values, GJR-t breaches too often at 99% and 99.5%.

Next steps: an expanding-window refit for all three models, filtered historical simulation, a machine-learning baseline, and USD/JPY or VIX as added explanatory variables.